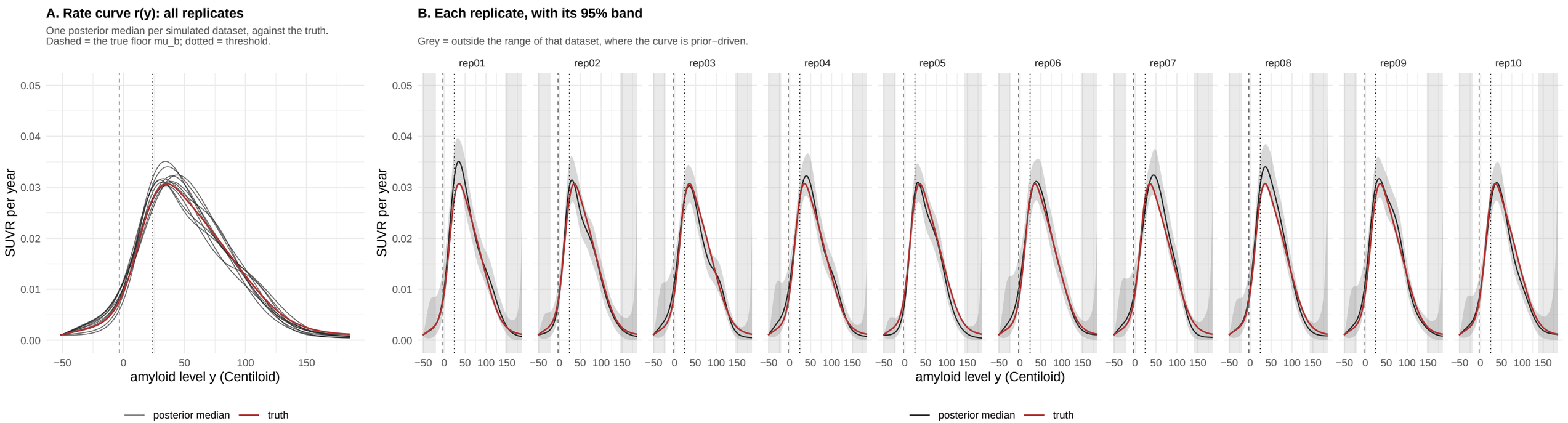
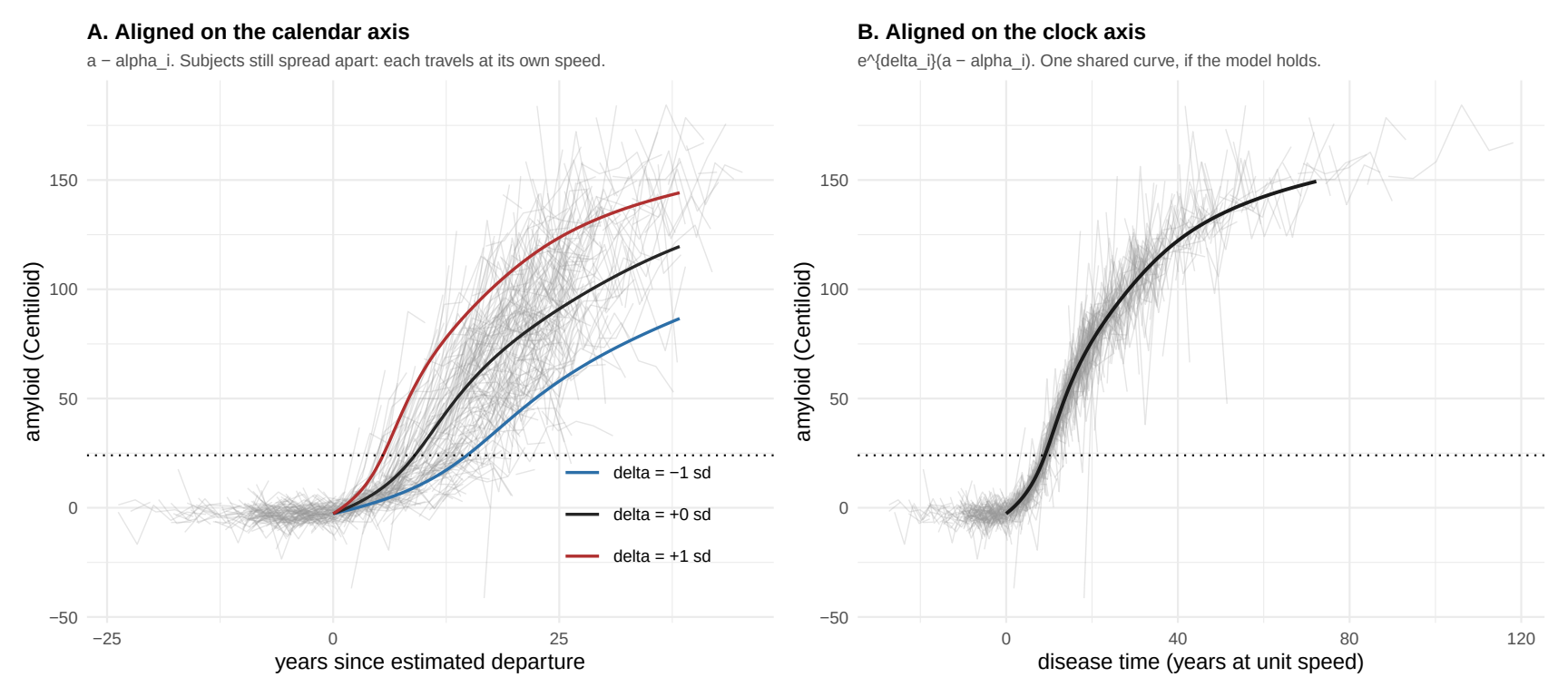
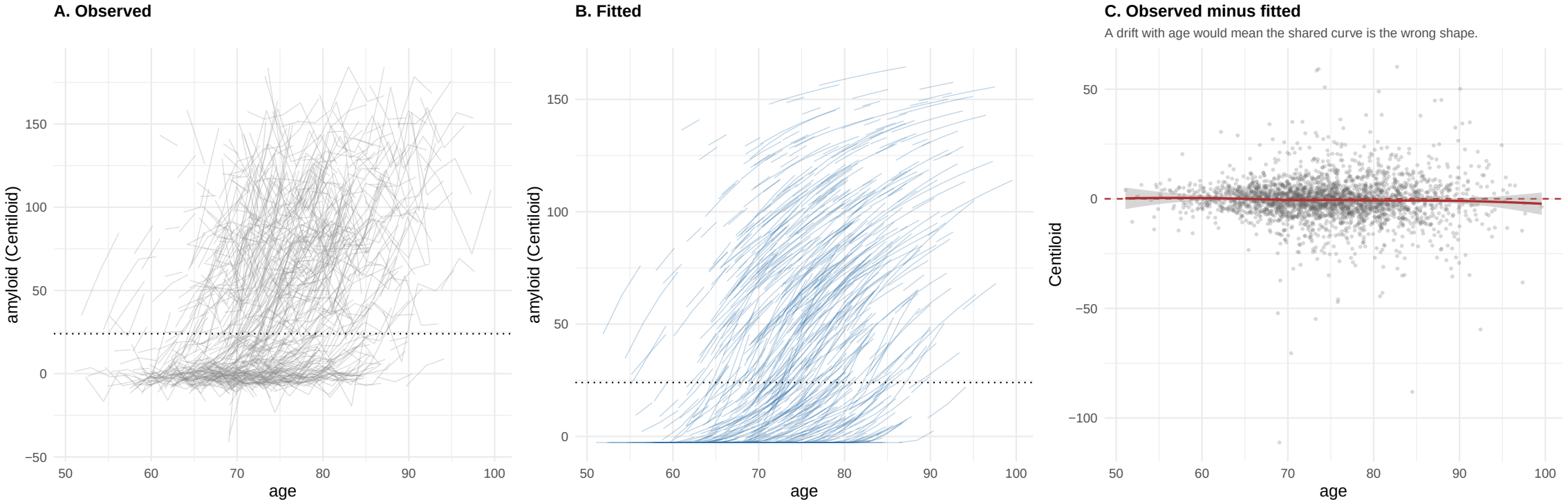
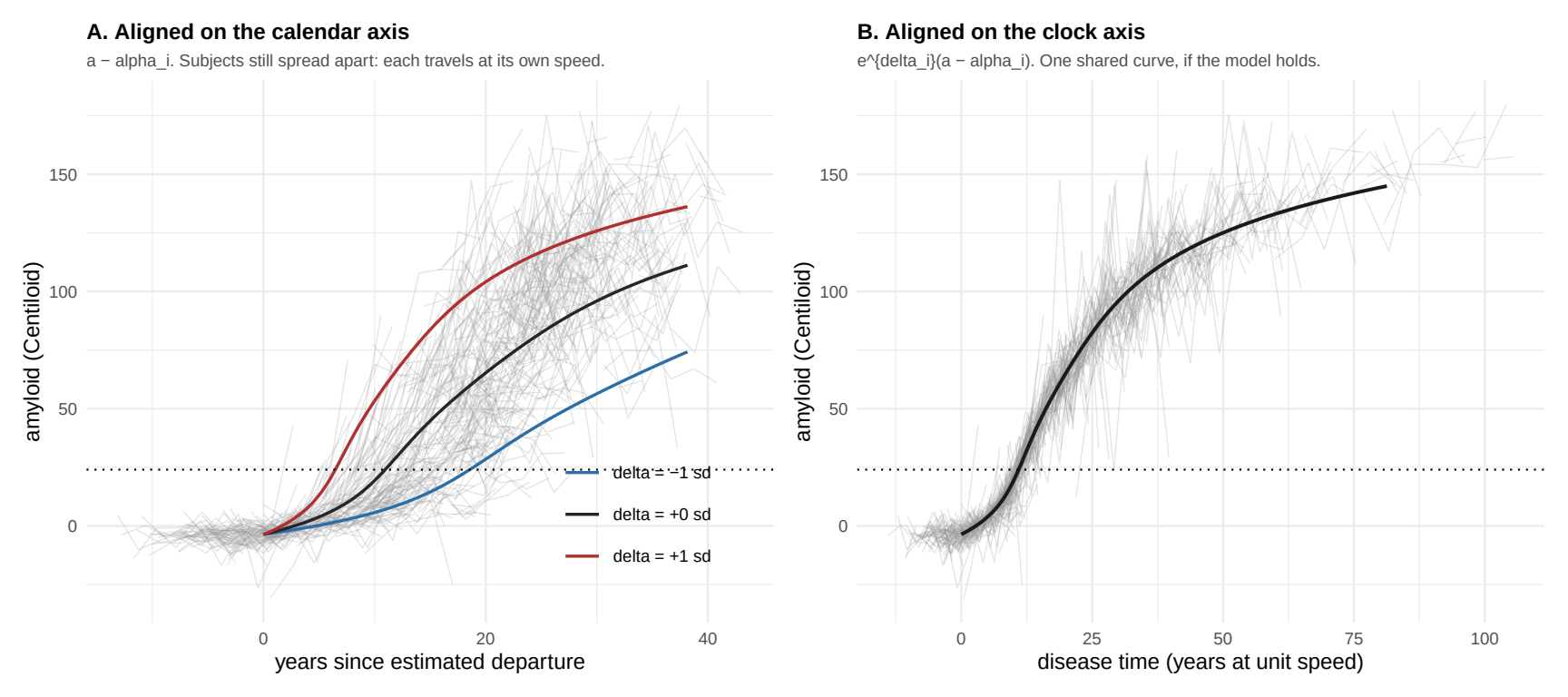
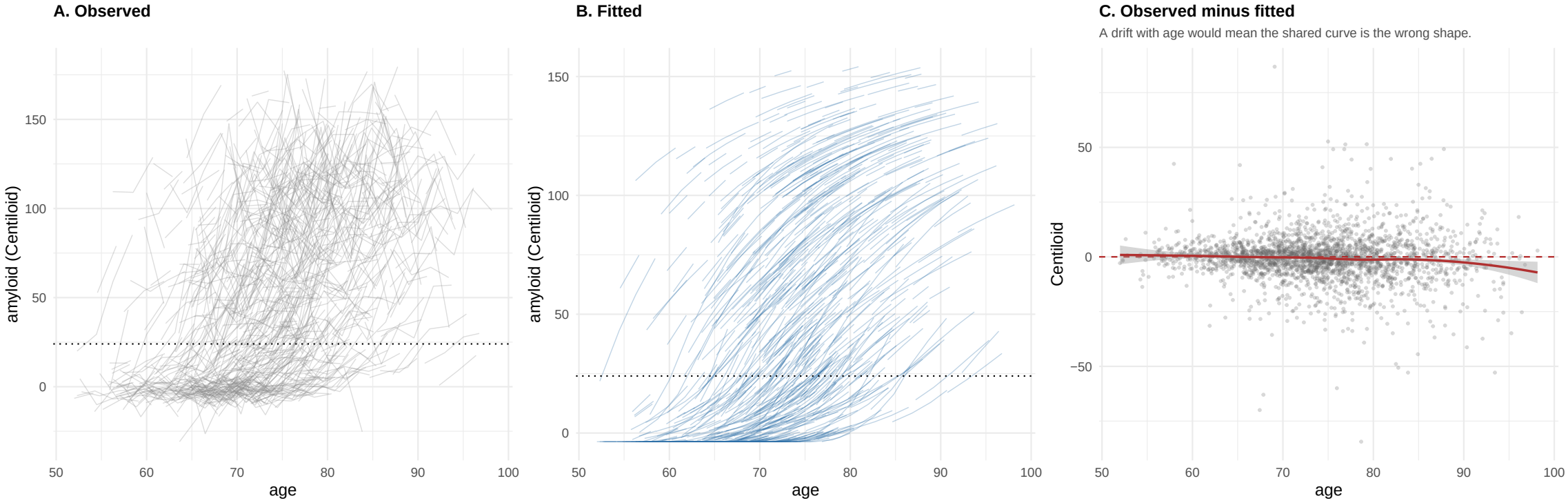
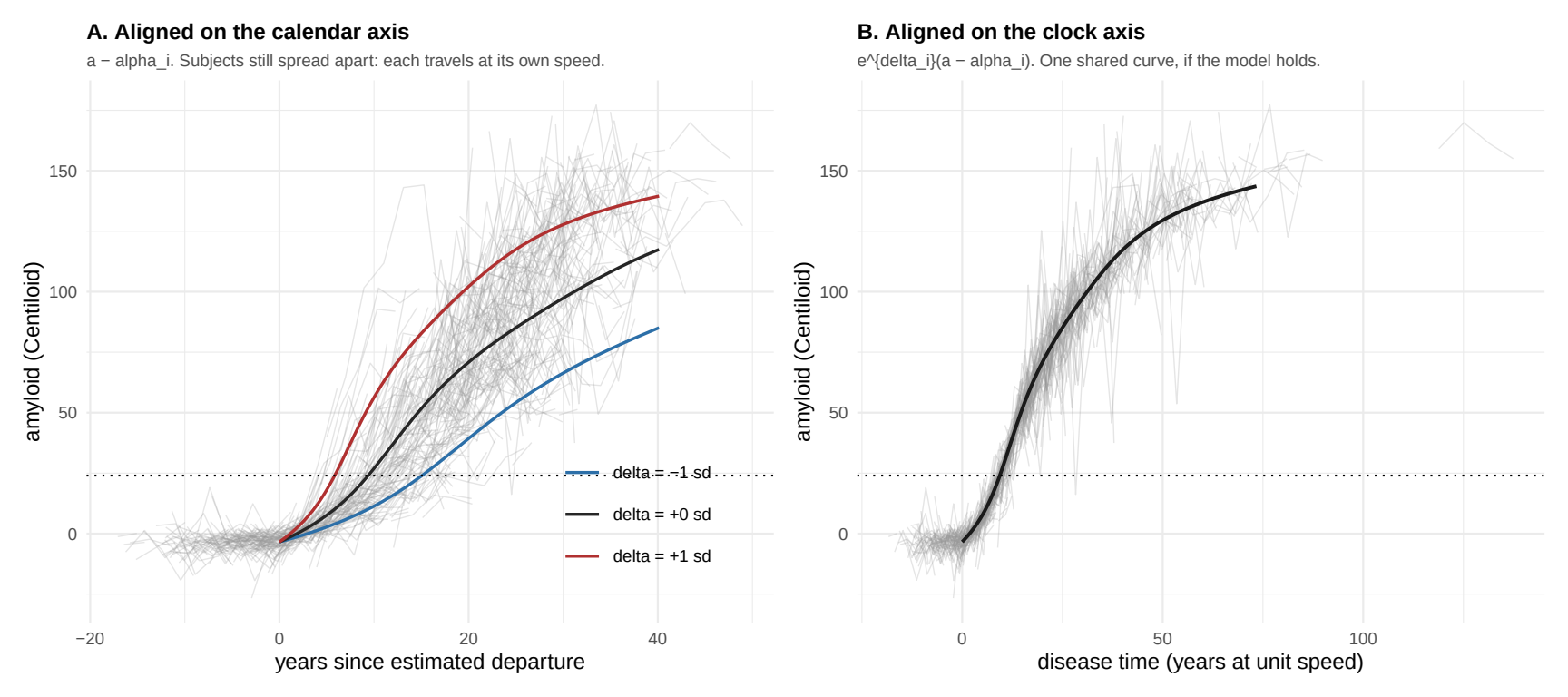

A. Rate curve r(y): all replicates
One posterior median per simulated dataset, against the truth.
Dashed = the true floor mu_b; dotted = threshold.
SUVR per year
amyloid level y (Centiloid)
posterior median
truth
B. Each replicate, with its 95% band
Grey = outside the range of that dataset, where the curve is prior−driven.
rep01
rep02
rep03
rep04
rep05
rep06
rep07
rep08
rep09
rep10
SUVR per year
amyloid level y (Centiloid)
posterior median
truth

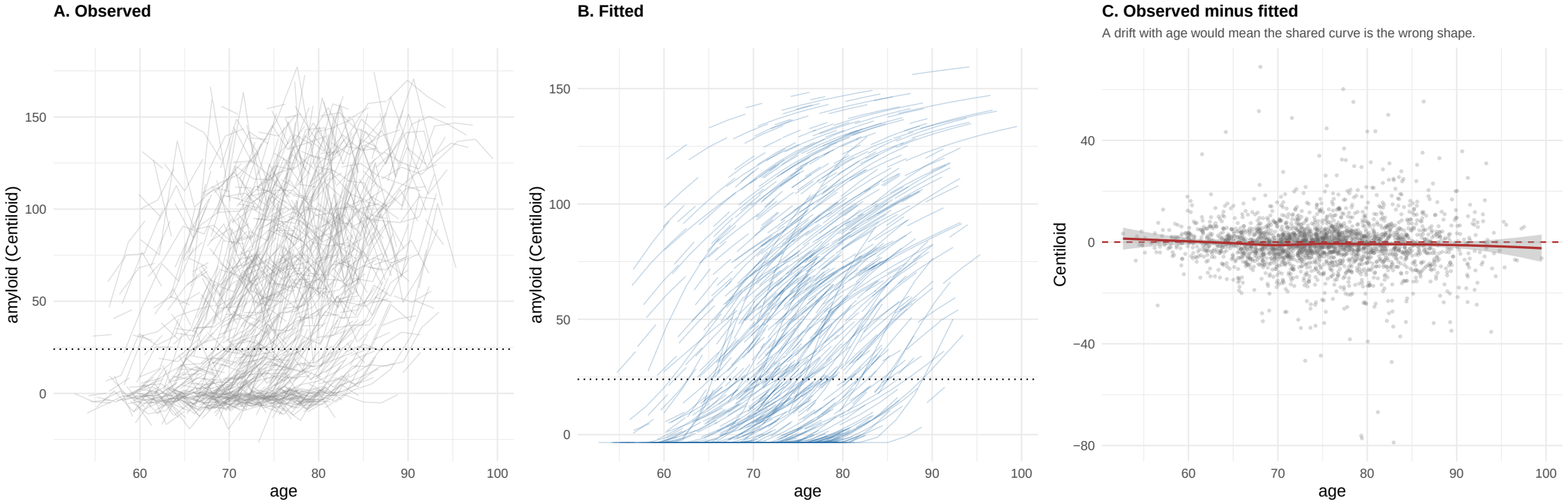

A. Aligned on the calendar axis
a − alpha_i. Subjects still spread apart: each travels at its own speed.
B. Aligned on the clock axis
e^{delta_i}(a − alpha_i). One shared curve, if the model holds.
amyloid (Centiloid)
150
100
50
0
−50
−25
0
25
years since estimated departure
delta = −1 sd
delta = +0 sd
delta = +1 sd
amyloid (Centiloid)
150
100
50
0
−50
0
40
80
120
disease time (years at unit speed)

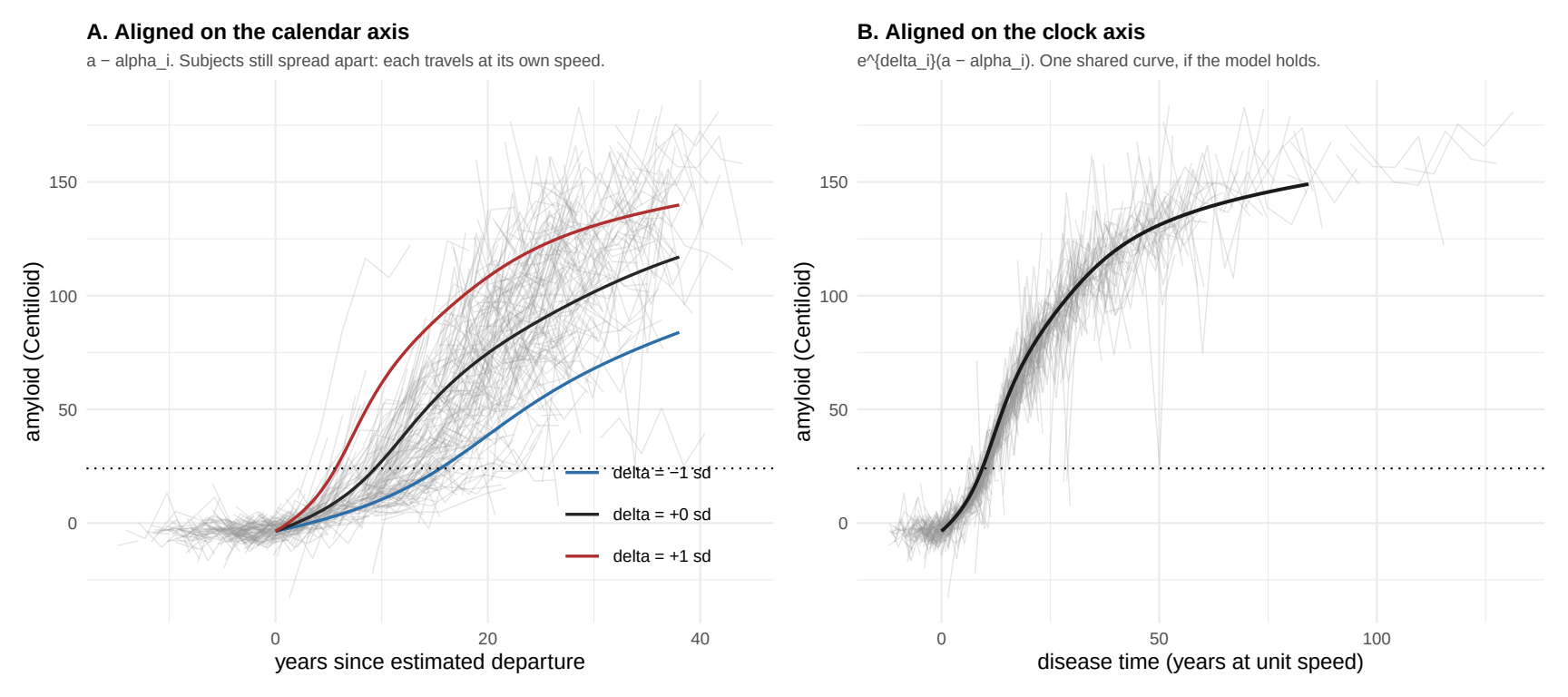

A. Observed
B. Fitted
C. Observed minus fitted
A drift with age would mean the shared curve is the wrong shape.
amyloid (Centiloid)
Centiloid
age
150
100
50
0
−50
−100
50
60
70
80
90
100

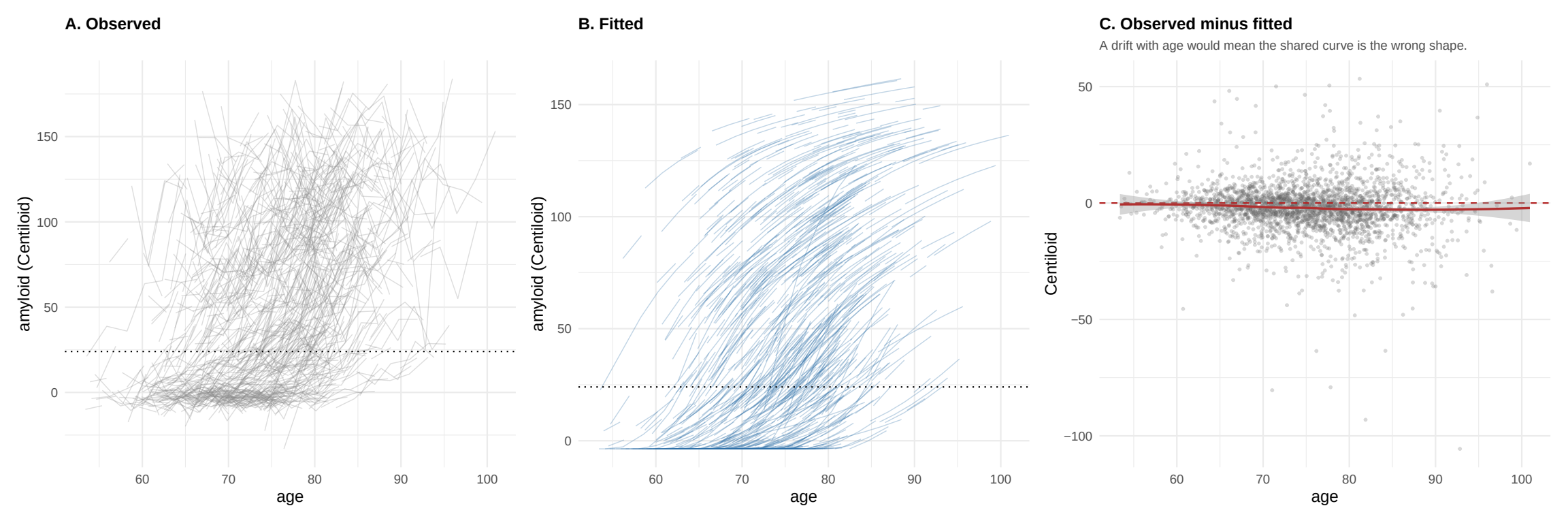

A. Aligned on the calendar axis
a − alpha_i. Subjects still spread apart: each travels at its own speed.
B. Aligned on the clock axis
e^{delta_i}(a − alpha_i). One shared curve, if the model holds.
amyloid (Centiloid)
150
100
50
0
delta = −1 sd
delta = +0 sd
delta = +1 sd
0
20
40
years since estimated departure
amyloid (Centiloid)
150
100
50
0
0
25
50
75
100
disease time (years at unit speed)

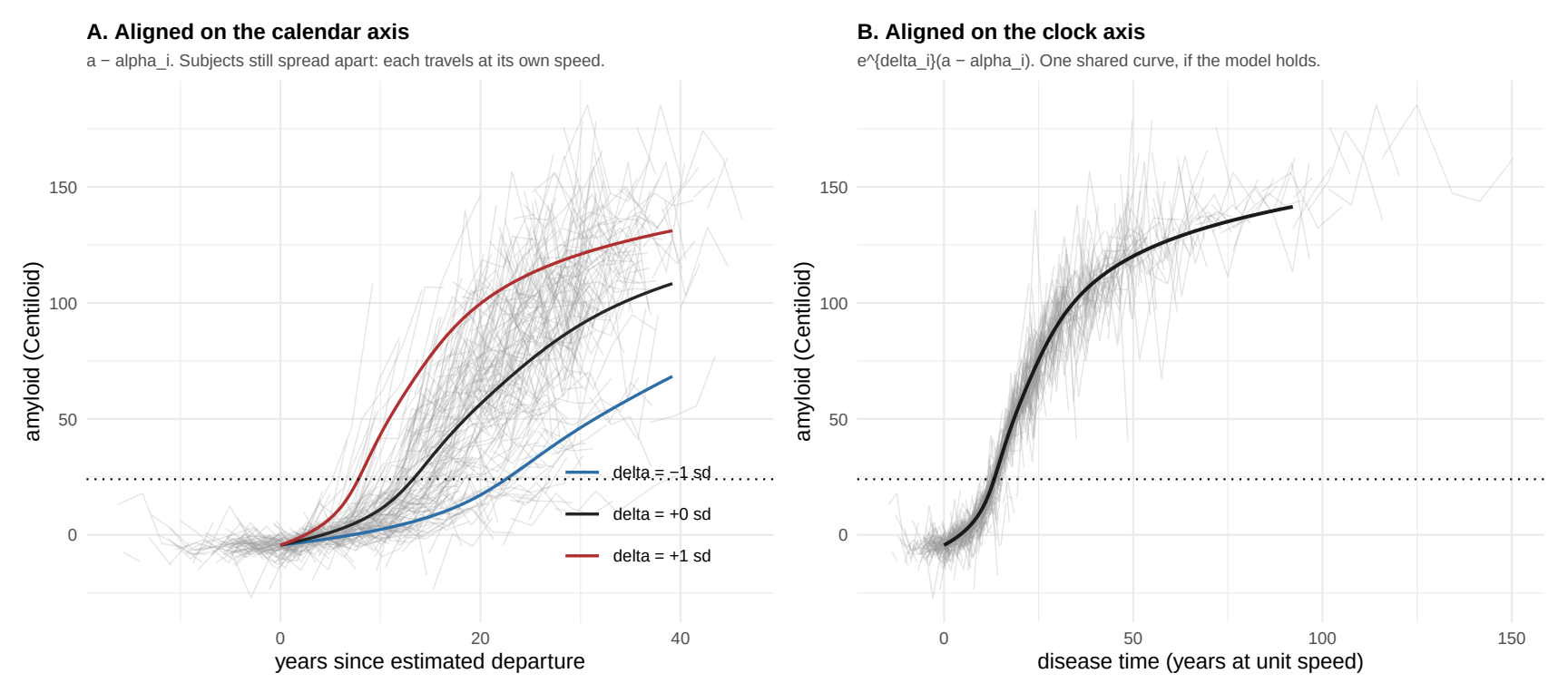

A. Observed
B. Fitted
C. Observed minus fitted
A drift with age would mean the shared curve is the wrong shape.
amyloid (Centiloid)
Centiloid
age
150
100
50
0
−50
50
60
70
80
90
100

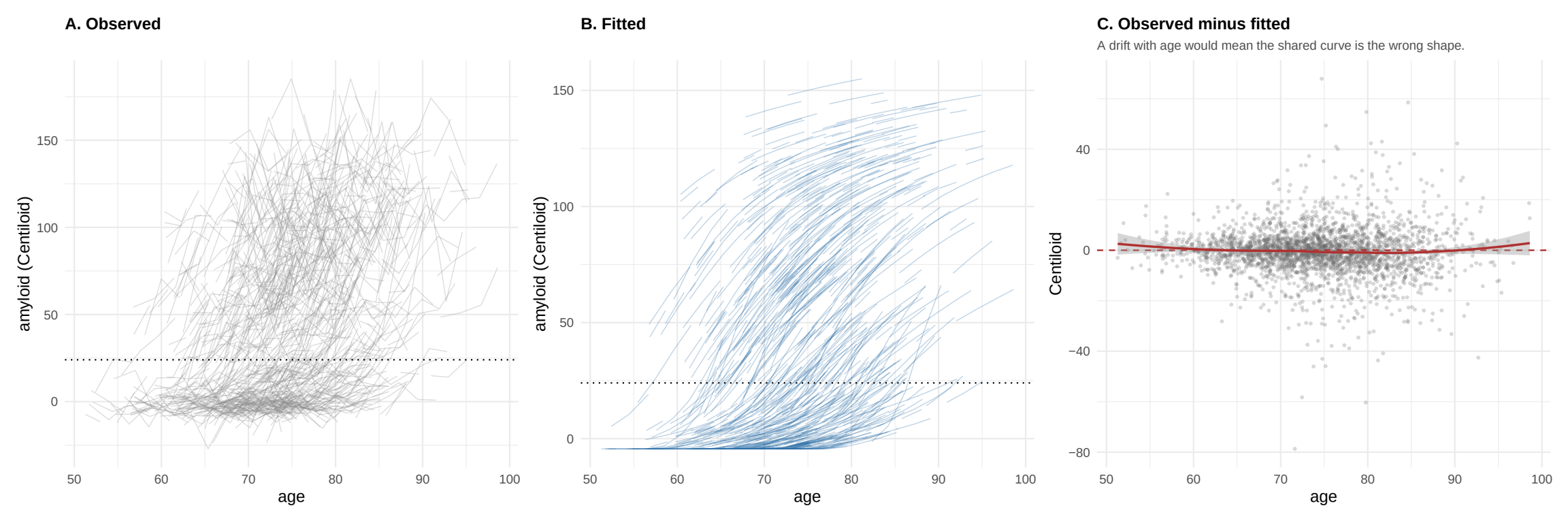

A. Aligned on the calendar axis
a − alpha_i. Subjects still spread apart: each travels at its own speed.
B. Aligned on the clock axis
e^{delta_i}(a − alpha_i). One shared curve, if the model holds.
amyloid (Centiloid)
150
100
50
0
−20
0
20
40
years since estimated departure
delta = −1 sd
delta = +0 sd
delta = +1 sd
amyloid (Centiloid)
150
100
50
0
0
50
100
disease time (years at unit speed)

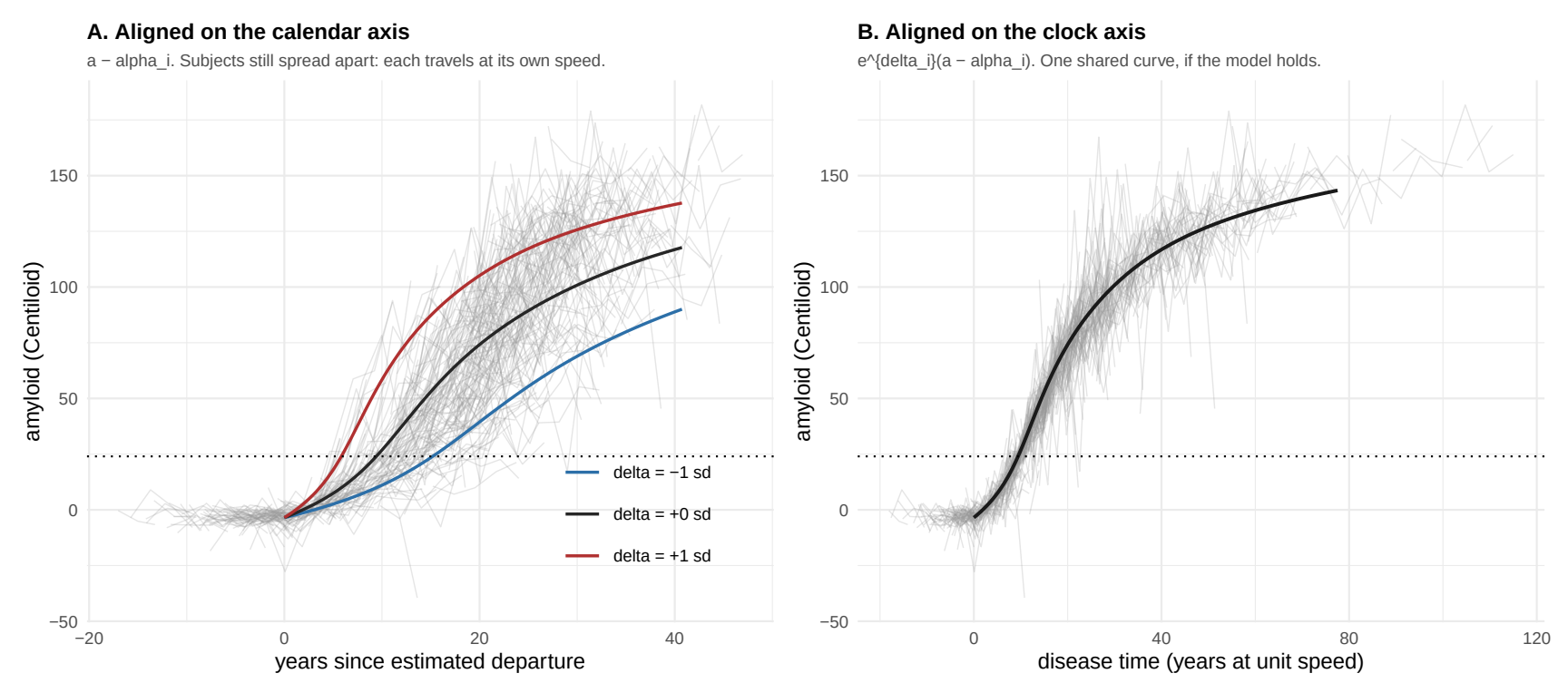

A. Observed
B. Fitted
C. Observed minus fitted
A drift with age would mean the shared curve is the wrong shape.
amyloid (Centiloid)
Centiloid
age
150
100
50
0
40
−40
−80
60
70
80
90
100

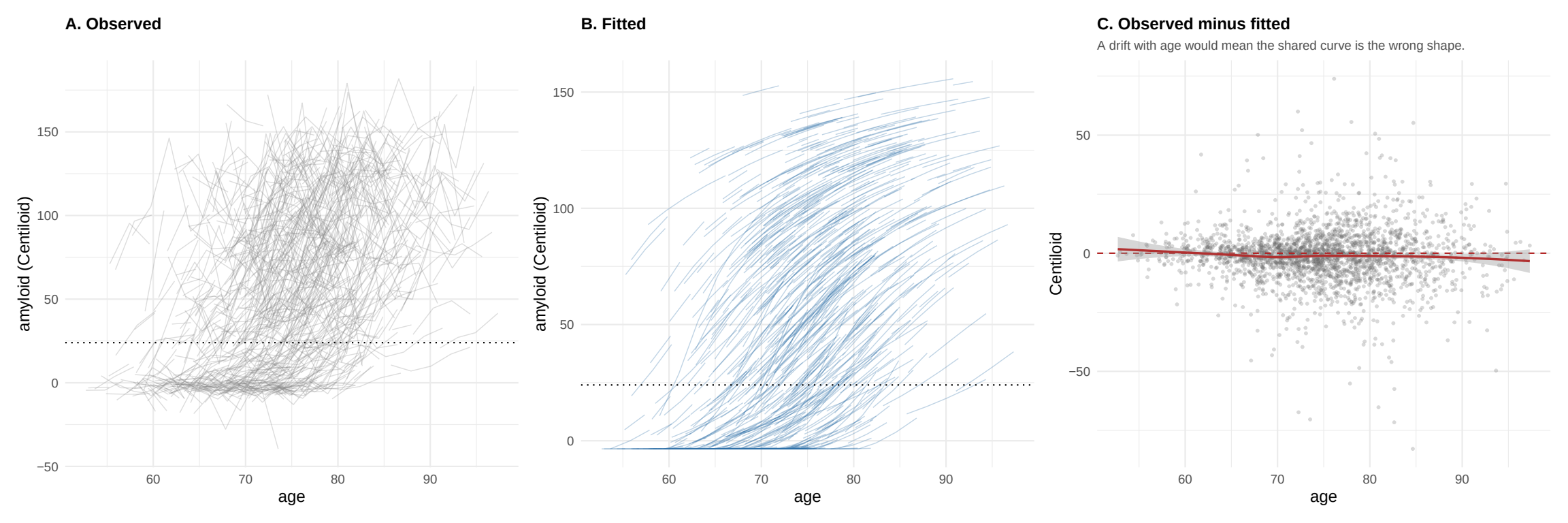

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a − alpha_i. Subjects still spread apart: each travels at its own speed.
B. Aligned on the clock axis
e^{delta_i}(a − alpha_i). One shared curve, if the model holds.
amyloid (Centiloid)
150
100
50
0
delta = −1 sd
delta = +0 sd
delta = +1 sd
0
20
40
years since estimated departure
amyloid (Centiloid)
150
100
50
0
0
50
100
disease time (years at unit speed)

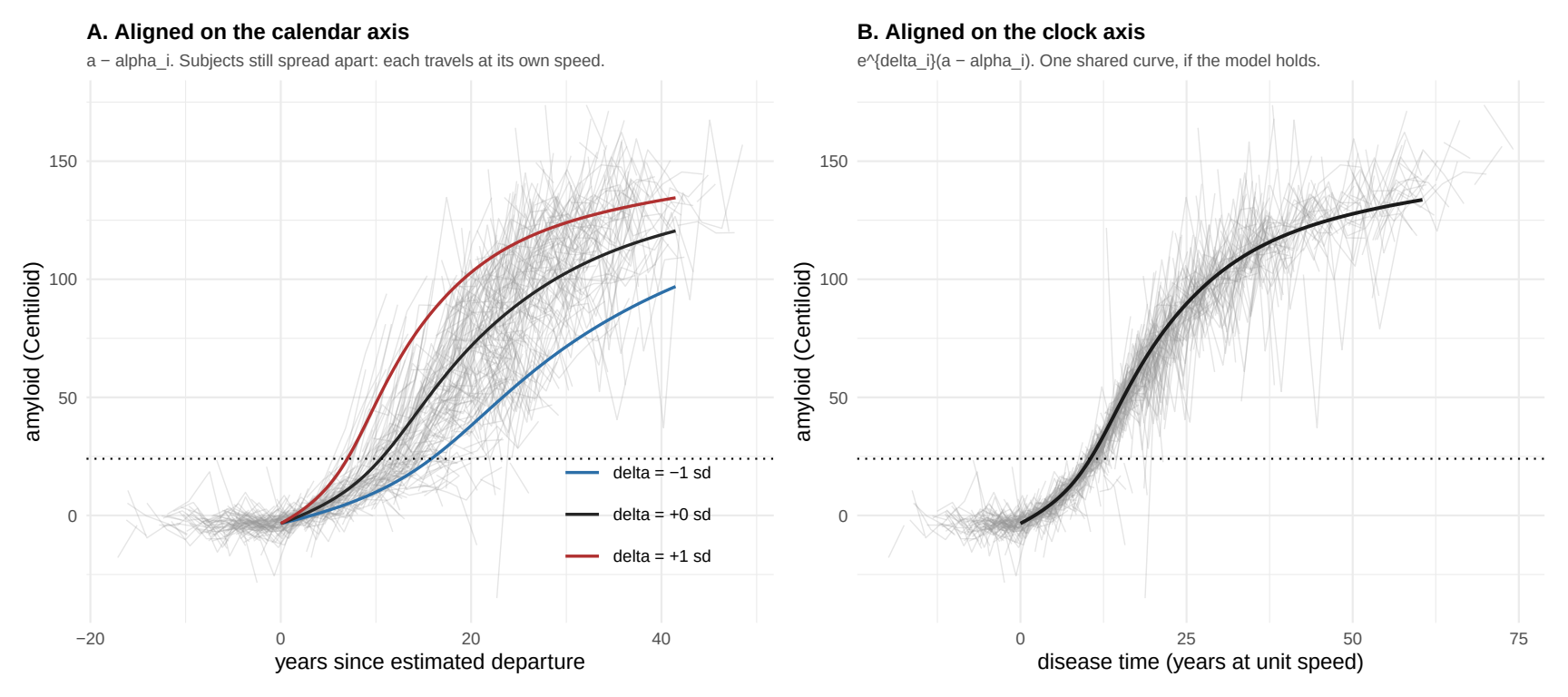

A. Observed
amyloid (Centiloid)
150
100
50
0
60
70
80
90
100
age
B. Fitted
amyloid (Centiloid)
150
100
50
0
60
70
80
90
100
age
C. Observed minus fitted
A drift with age would mean the shared curve is the wrong shape.
Centiloid
50
0
−50
−100
60
70
80
90
100
age

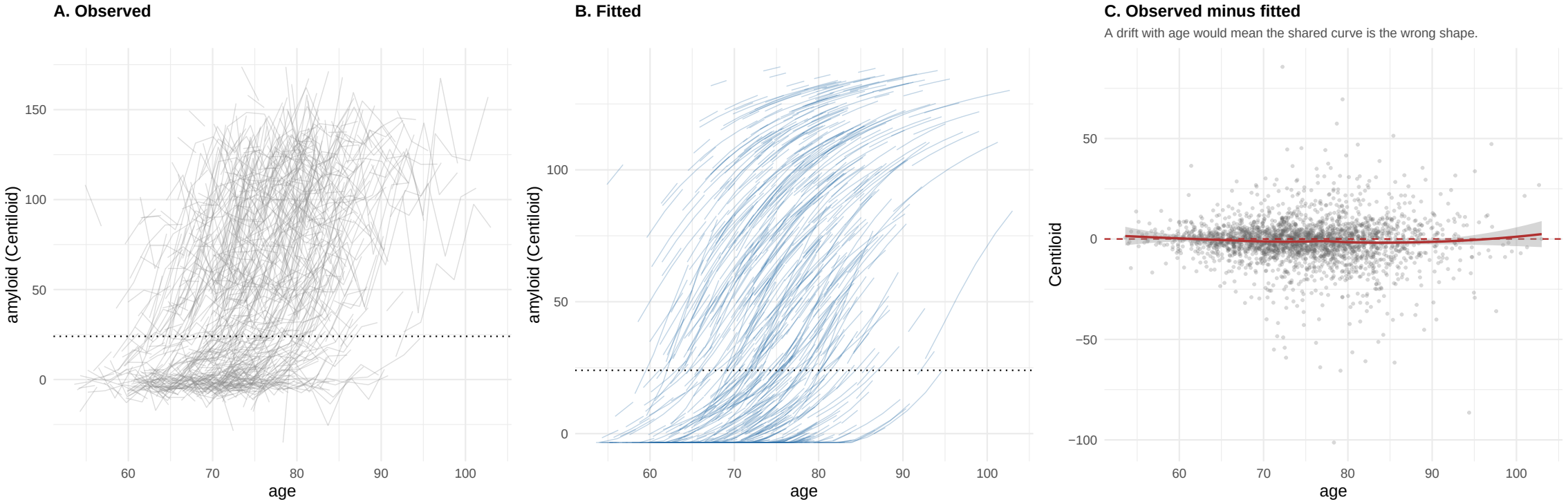

A. Aligned on the calendar axis
a − alpha_i. Subjects still spread apart: each travels at its own speed.
amyloid (Centiloid)
150
100
50
0
0
20
40
years since estimated departure
delta = −1 sd
delta = +0 sd
delta = +1 sd
B. Aligned on the clock axis
e^{delta_i}(a − alpha_i). One shared curve, if the model holds.
amyloid (Centiloid)
150
100
50
0
0
50
100
150
disease time (years at unit speed)

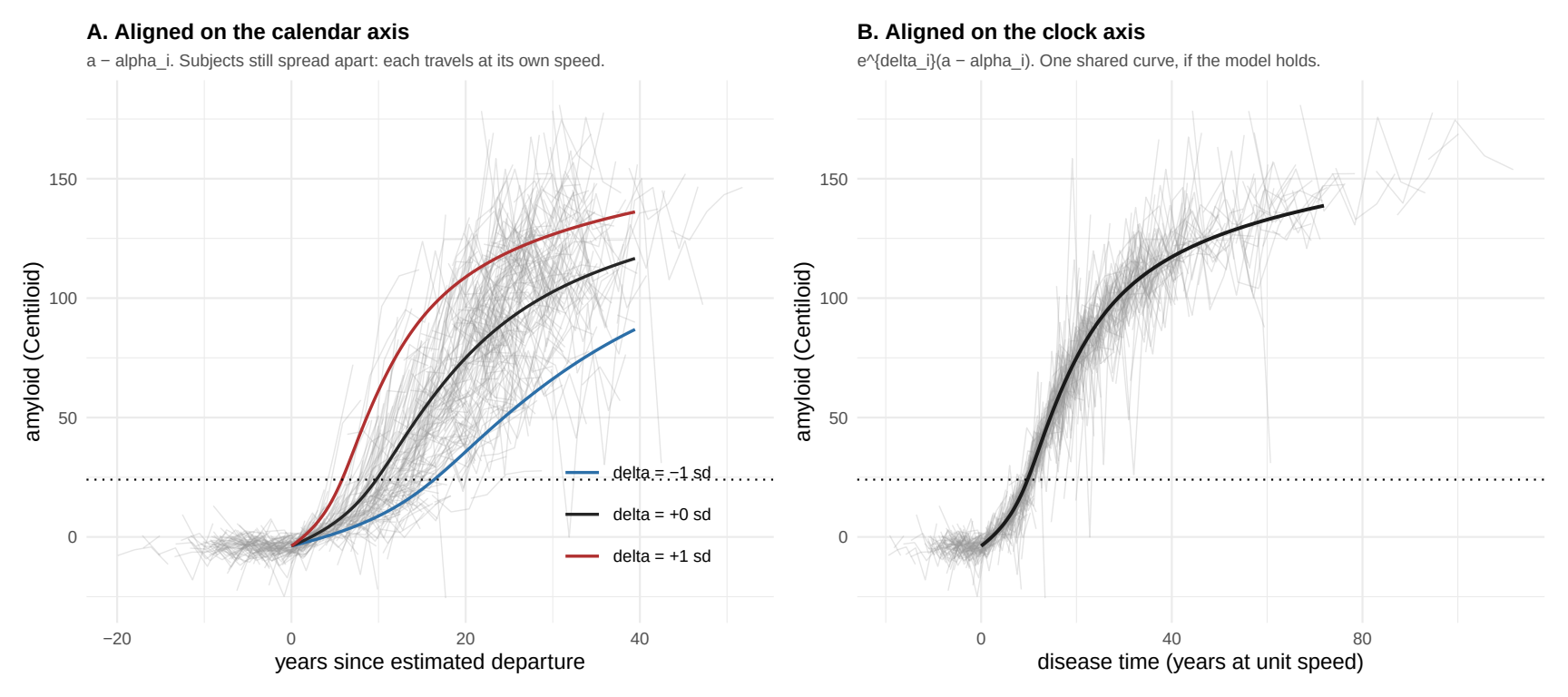

A. Observed
amyloid (Centiloid)
150
100
50
0
50
60
70
80
90
100
age
B. Fitted
amyloid (Centiloid)
150
100
50
0
50
60
70
80
90
100
age
C. Observed minus fitted
A drift with age would mean the shared curve is the wrong shape.
Centiloid
40
0
−40
−80
50
60
70
80
90
100
age

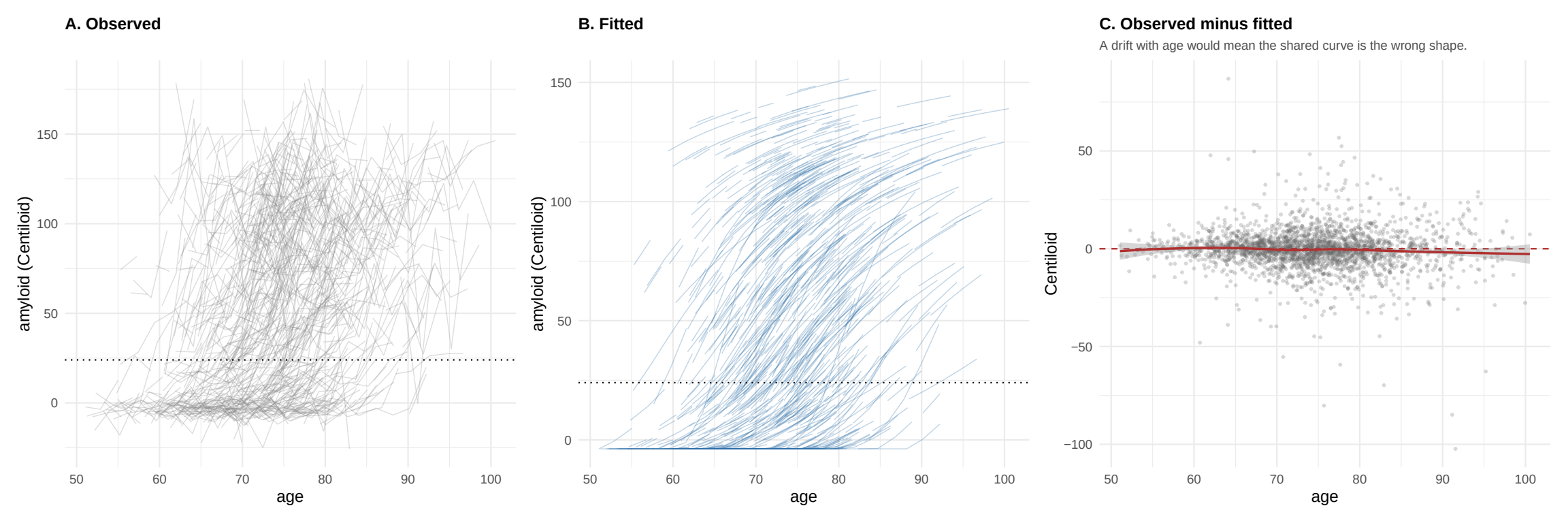

A. Aligned on the calendar axis
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e^{delta_i}(a − alpha_i). One shared curve, if the model holds.
amyloid (Centiloid)
150
100
50
0
−50
−20
0
20
40
years since estimated departure
delta = −1 sd
delta = +0 sd
delta = +1 sd
amyloid (Centiloid)
150
100
50
0
−50
0
40
80
120
disease time (years at unit speed)

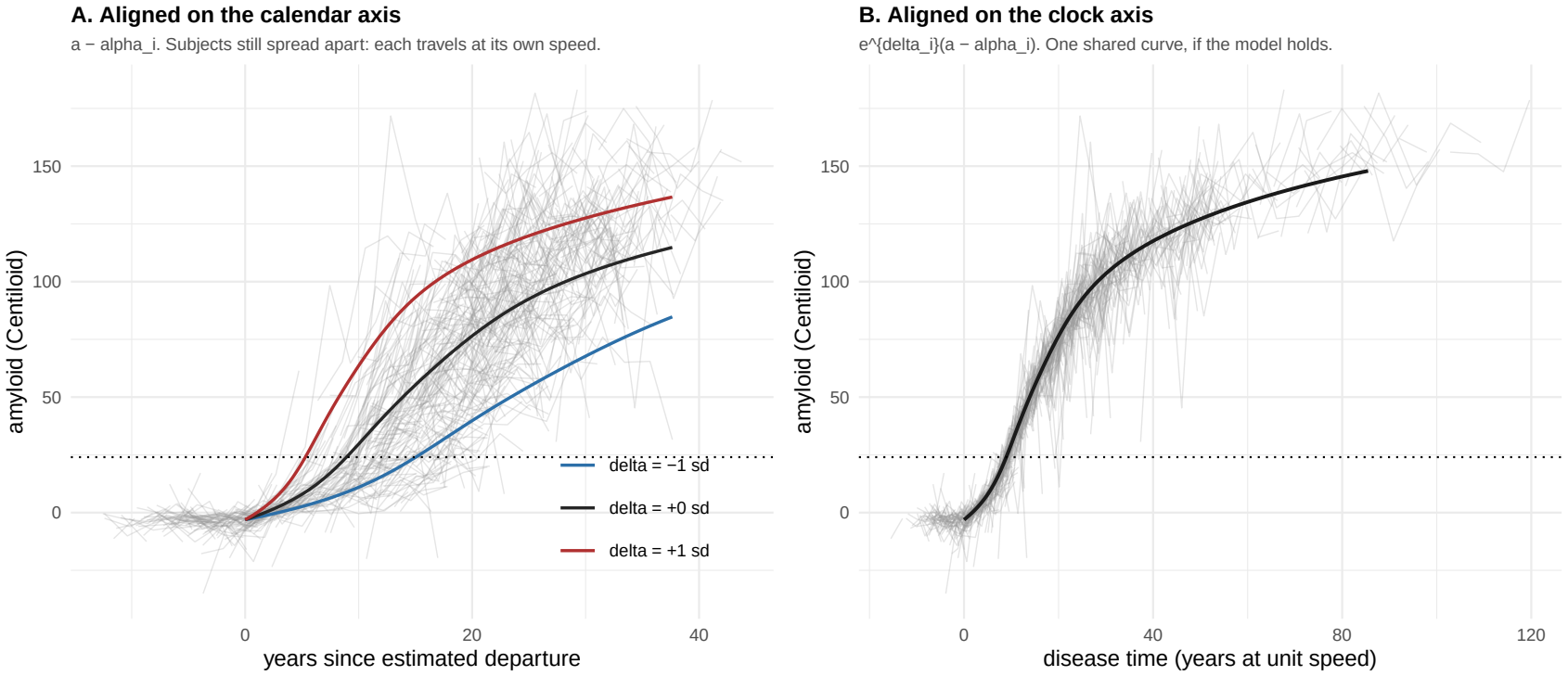

A. Observed
amyloid (Centiloid)
150
100
50
0
−50
60
70
80
90
age
B. Fitted
amyloid (Centiloid)
150
100
50
0
60
70
80
90
age
C. Observed minus fitted
A drift with age would mean the shared curve is the wrong shape.
Centiloid
50
0
−50
60
70
80
90
age

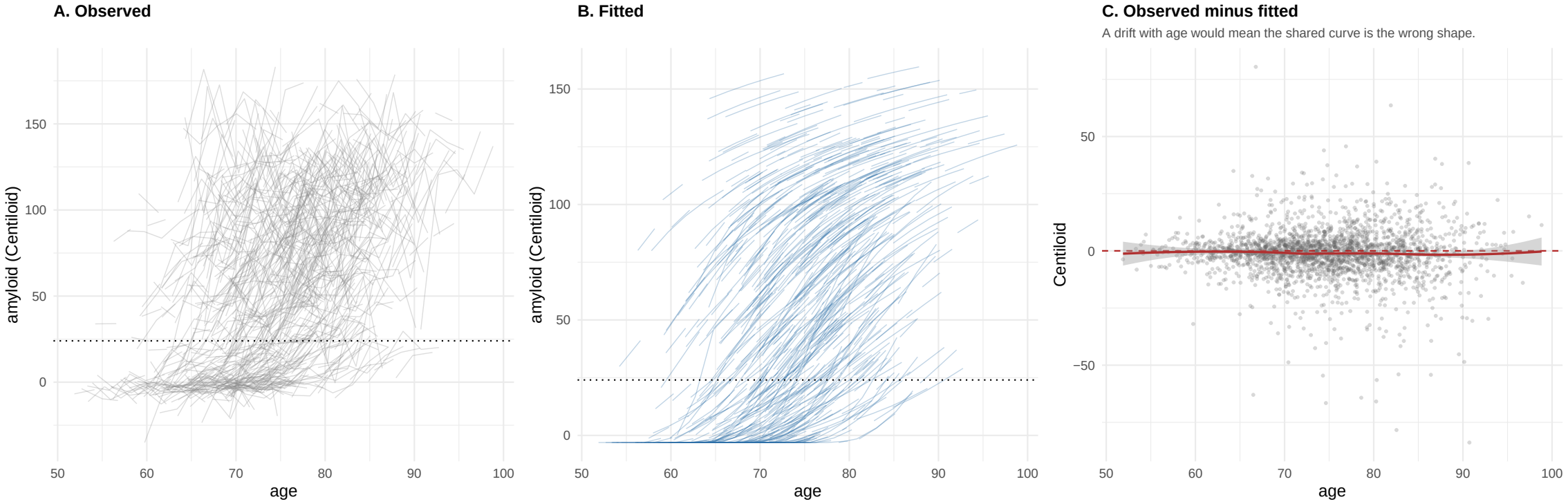

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years since estimated departure
delta = −1 sd
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delta = +1 sd
B. Aligned on the clock axis
e^{delta_i}(a − alpha_i). One shared curve, if the model holds.
amyloid (Centiloid)
150
100
50
0
0
25
50
75
disease time (years at unit speed)

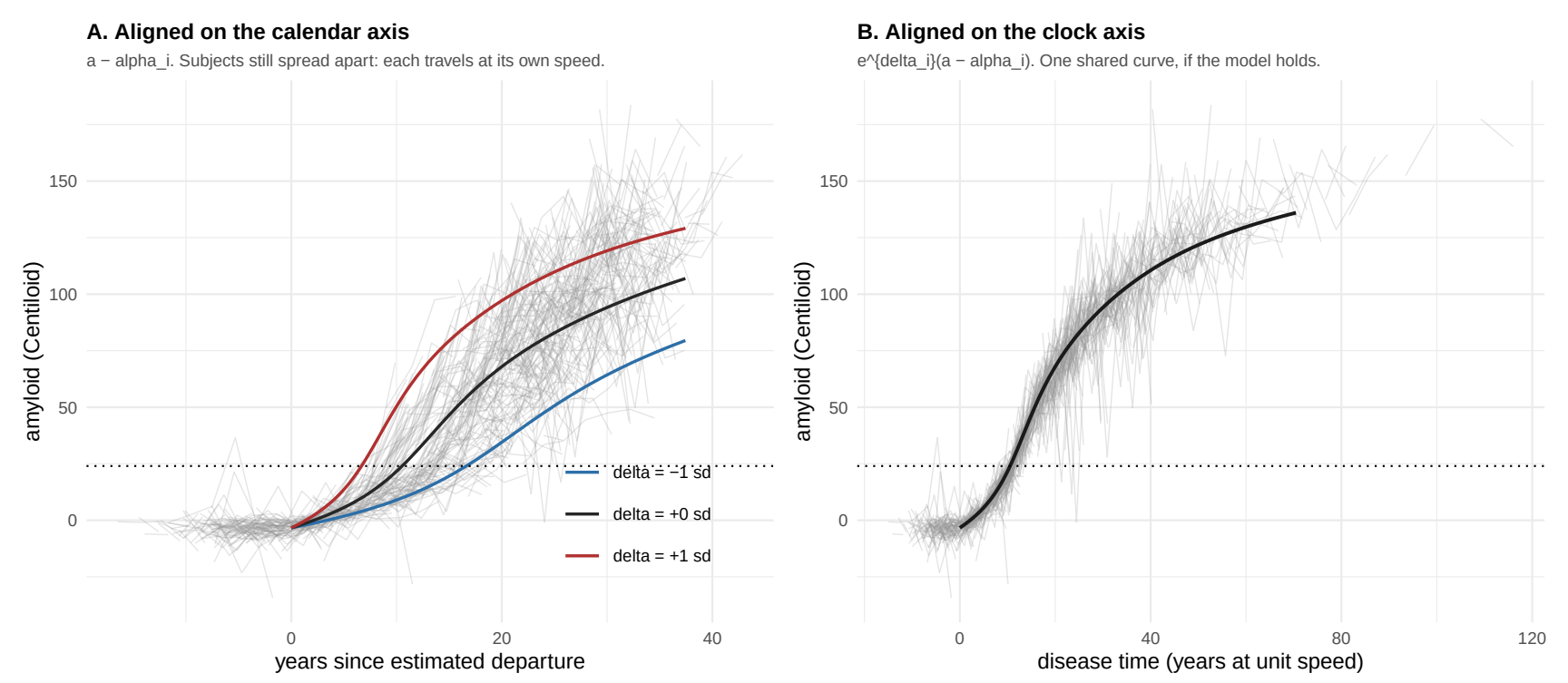

A. Observed
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C. Observed minus fitted
A drift with age would mean the shared curve is the wrong shape.
amyloid (Centiloid)
Centiloid
age
150
100
50
0
−50
−100
60
70
80
90
100

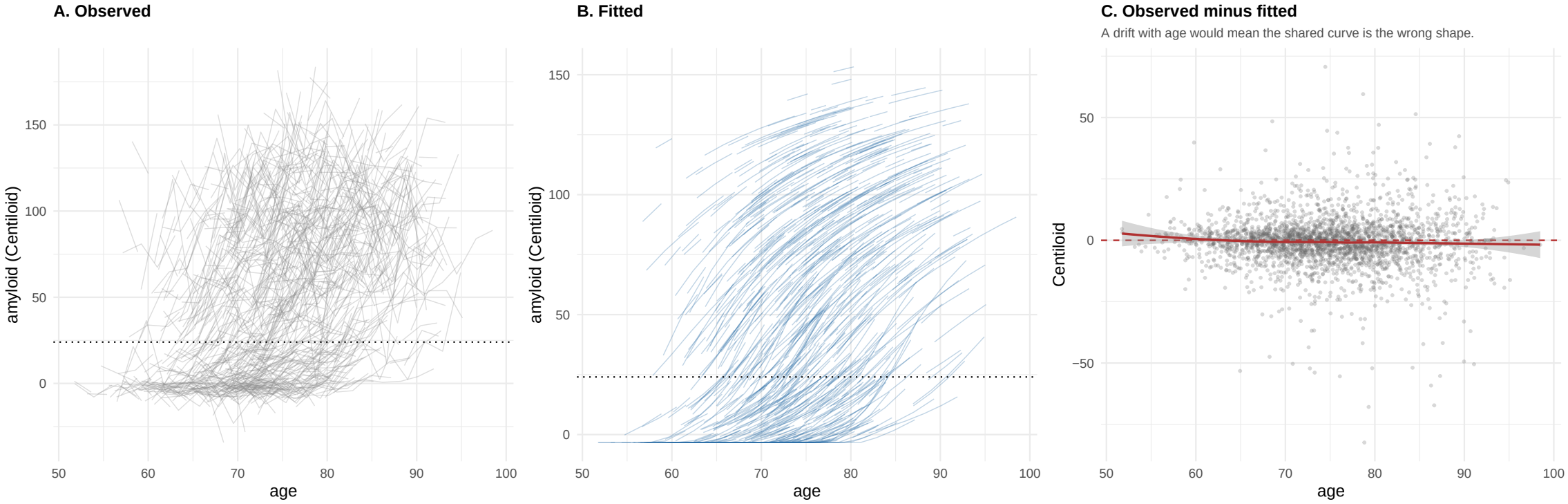

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delta = +0 sd
delta = +1 sd
amyloid (Centiloid)
150
100
50
0
0
40
80
disease time (years at unit speed)


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C. Observed minus fitted
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Centiloid
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50
60
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80
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100


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delta = +1 sd
B. Aligned on the clock axis
e^{delta_i}(a − alpha_i). One shared curve, if the model holds.
amyloid (Centiloid)
150
100
50
0
0
40
80
120
disease time (years at unit speed)


A. Observed
B. Fitted
C. Observed minus fitted
A drift with age would mean the shared curve is the wrong shape.
amyloid (Centiloid)
Centiloid
age
150
100
50
0
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60
70
80
90
100


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delta = −1 sd
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delta = +1 sd
0
20
40
years since estimated departure
amyloid (Centiloid)
150
100
50
0
0
40
80
120
disease time (years at unit speed)


A. Observed
B. Fitted
C. Observed minus fitted
A drift with age would mean the shared curve is the wrong shape.
amyloid (Centiloid)
Centiloid
age
150
100
50
0
−50
50
60
70
80
90
100